Math 5324H Lec16

Suppose the wave equation

IVP $\begin{cases} u_{tt} - c^2 u_{xx} = 0, & x \in \mathbb{R},\ t > 0 \\ u(x,0) = \phi(x), & x \in \mathbb{R} \\ u_t(x,0) = \psi(x), & x \in \mathbb{R} \end{cases}$

Energy: $E[u] = \frac{1}{2}\int_{\mathbb{R}} |u_t|^2 + |u_x|^2 \, dx$

We have already prove $\frac{dE}{dt} = 0 \Rightarrow \underline{E[u](t) = E[u](0)}$

Uniqueness: u_1 and u_2 satified IVP

$w = u_1 - u_2$

$\begin{cases} w_{tt} - c^2 w_{xx} = 0, & x \in \mathbb{R},\ t > 0 \\ w(x,0) = \phi_1 - \phi_2 = 0, & x \in \mathbb{R} \\ w_t(x,0) = \psi_1 - \psi_2 = 0, & x \in \mathbb{R} \end{cases}$

$\underline{0 \le E[w](t) = E[w](0)}$

$E[u](0) = \frac{1}{2}\int_{\mathbb{R}} |\underset{0}{w_t}|^2 + |\underset{0}{w_x}|^2 \, dx$

$\underline{0 \le E[w](t)} = \underline{E[w](0) = 0}$

$\Downarrow$

$w_t(x,t) = 0 \Rightarrow \cancel{w(x,t) = x + C}$

$w_x(x,t) = 0 \Rightarrow \cancel{w(x,t) = t + C}$

$w(x,t)$ is constant in time

$w(x,t)$ is constant in x

$\boxed{w(x,t) = \text{constant}} \xrightarrow{w(x,0) = 0} \boxed{w(x,t) = 0}$

$u_1 = u_2 \quad \forall (x,t)$

property:

① ~~Sa~~ Scaling $\lambda > 0$

$u = (\sqrt{\lambda}x, \lambda t)$ is also solution

~~② $u(-x, t)$ is also solution~~

③ $u(x-y, t)$ is also solution

④ Any derivative of u is also solution

④ $\int_R u(x-y, t) g(y)\, dy$ is also solution

$$H(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases} \qquad H(\lambda x) = H(x)$$

$$Q(x, 0) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\begin{cases} Q_t - k Q_{xx} = 0, & x \in R,\ t > 0 \\ Q(x, 0) = H(x), & x \in R \end{cases}$$

$Q(\sqrt{\lambda}x, \lambda t)$ is also solution

self-semitor variable

$$Q(x, t) = g\left(\frac{x}{\sqrt{t}}\right)$$

$g(\cancel{\sqrt{4k}}\, z) = g(\sqrt{4k} z)$

$$= g\left(\frac{x}{\sqrt{4kt}}\right)$$

$$= g(p) \qquad p: \frac{x}{\sqrt{4kt}}$$

$$Q_t = -\frac{1}{2t}\frac{x}{\sqrt{4kt}}g'(p)$$

$$Q_x = \frac{1}{\sqrt{4kt}}g'(p)$$

$$Q_{xx} = \frac{1}{4kt}g''(p)$$

$$0 = Q_t - kQ_{xx} = \frac{1}{t}\left[-\frac{1}{2}pg'(p) - \frac{1}{4}g''(p)\right]$$

$$\Downarrow$$

$$\boxed{g'' + 2pg' = 0} \quad (g = g(p))$$

$$Q(x,t) = g(p) = C_1\int e^{-p^2}dp + C_2$$

$$= C_1\int^{x/\sqrt{4kt}} e^{-p^2}dp + C_2 \quad (t>0)$$

if $x \geq 0$ $\quad 1 = \lim_{t\to 0^+} Q(x,t) = C_1\int_0^{\infty} e^{-p^2}dp + C_2$

$$= C_1\frac{\sqrt{\pi}}{2} + C_2$$

if $x<0$, $\quad 0 = \lim_{t\to 0^+} Q(x,t) = C_1\int_0^{-\infty} e^{-p^2}dp + C_2$

$$= -C_1\frac{\sqrt{\pi}}{2} + C_2$$

$$\Downarrow$$

$$\begin{cases} C_2 = \frac{1}{2} \\ C_1\sqrt{\pi} = 1 \Rightarrow C_1 = \frac{1}{\sqrt{\pi}} \end{cases}$$

$$Q(x,t) = \frac{1}{\sqrt{\pi}} + \frac{1}{2}\int^{x/\sqrt{4kt}} e^{-p^2}dp \quad (t>0)$$